Retail Sales Data Analysis Using Python

Objective

The objective of this project is to analyze retail sales data and identify sales trends, customer behavior, and business insights using Python.

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```
df=pd.read_csv("retail_sales_dataset.csv")
```

In [6]:
```
df.head()
```

Out[6]:

	transaction_id	transaction_date	customer_id	customer_gender	customer_age_group
0	T0000001	2024-04-24	C000820	Other	35-44
1	T0000002	2025-07-12	C002849	Other	45-54
2	T0000003	2025-06-01	C019727	Male	55+
3	T0000004	2025-08-26	C009116	Male	25-34
4	T0000005	2024-12-10	C003350	Male	45-54

In [8]:
```
df.shape
```

Out[8]:
```
(120000, 17)
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 120000 entries, 0 to 119999
Data columns (total 17 columns):
 #   Column             Non-Null Count   Dtype
---  ------             --------------   -----
 0   transaction_id     120000 non-null  str
 1   transaction_date   120000 non-null  str
 2   customer_id        120000 non-null  str
 3   customer_gender    120000 non-null  str
 4   customer_age_group 120000 non-null  str
 5   customer_segment   120000 non-null  str
 6   product_id         120000 non-null  str
 7   product_name       120000 non-null  str
 8   category           120000 non-null  str
 9   brand              120000 non-null  str
 10  quantity           120000 non-null  int64
 11  unit_price         120000 non-null  float64
 12  discount_pct       120000 non-null  int64
 13  sales_amount       120000 non-null  float64
 14  payment_method     120000 non-null  str
 15  sales_channel      120000 non-null  str
 16  region             120000 non-null  str
dtypes: float64(2), int64(2), str(13)
memory usage: 15.6 MB
```

In [12]:

```
df.isnull().sum()
```

Out[12]:

```
transaction_id        0
transaction_date      0
customer_id           0
customer_gender       0
customer_age_group    0
customer_segment      0
product_id            0
product_name          0
category              0
brand                 0
quantity              0
unit_price            0
discount_pct          0
sales_amount          0
payment_method        0
sales_channel         0
region                0
dtype: int64
```

In [14]:

```
df.describe()
```

Out[14]:

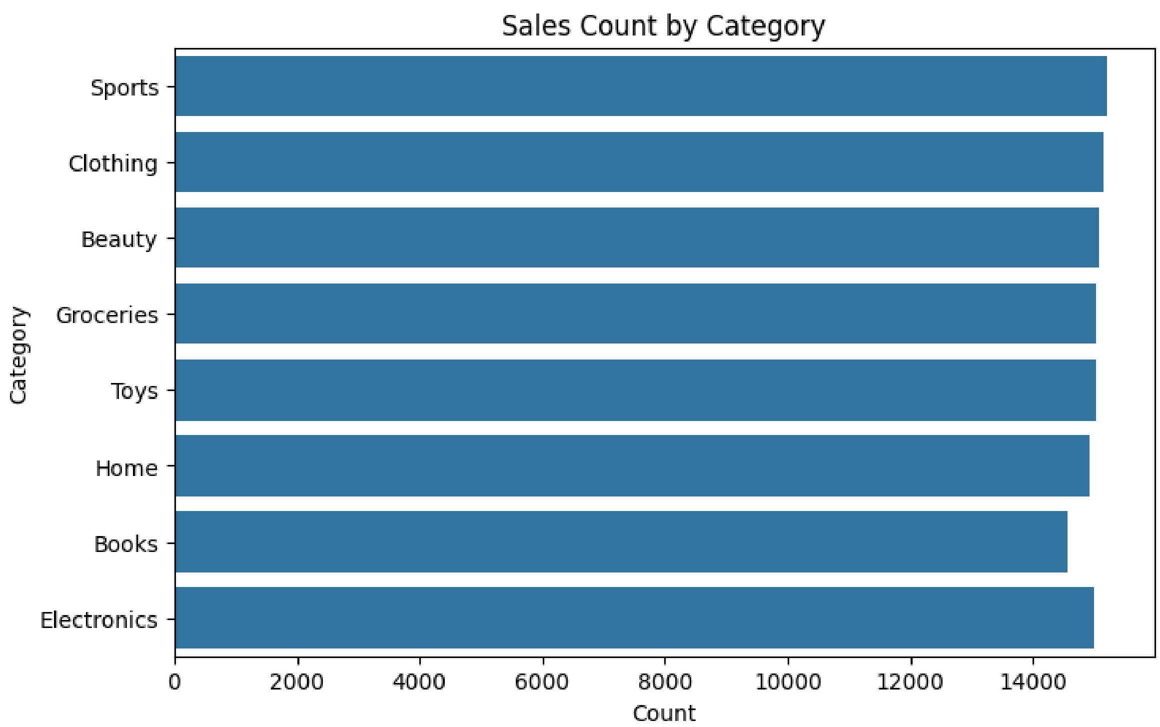
	quantity	unit_price	discount_pct	sales_amount
count	120000.000000	120000.000000	120000.000000	120000.000000
mean	1.662908	240.621785	5.496500	377.975454
std	1.014291	146.457057	8.193257	356.893357
min	1.000000	7.730000	0.000000	5.410000
25%	1.000000	102.010000	0.000000	136.860000
50%	1.000000	238.750000	0.000000	295.980000
75%	2.000000	379.000000	10.000000	461.500000
max	5.000000	493.510000	30.000000	2467.550000

In [16]:
```
df.columns
```

Out[16]:
```
Index(['transaction_id', 'transaction_date', 'customer_id', 'customer_gender',
       'customer_age_group', 'customer_segment', 'product_id', 'product_name',
       'category', 'brand', 'quantity', 'unit_price', 'discount_pct',
       'sales_amount', 'payment_method', 'sales_channel', 'region'],
      dtype='str')
```

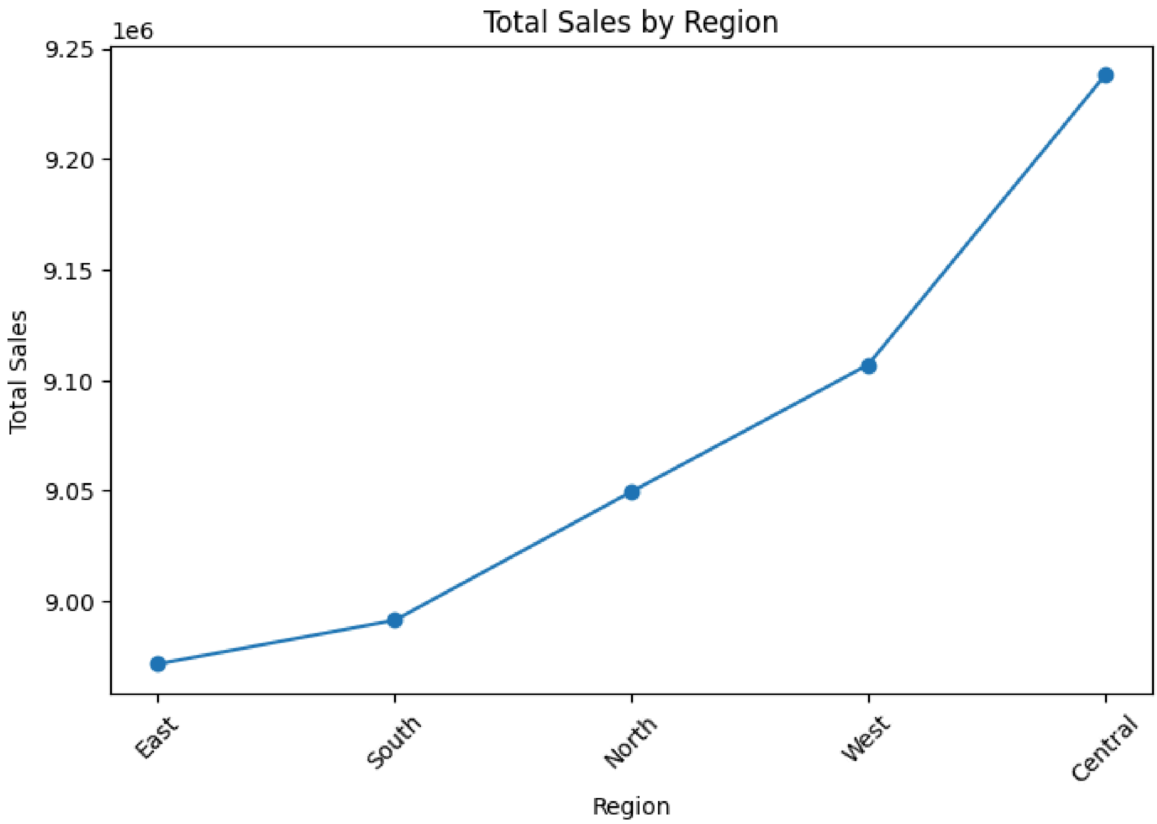
Sales Count by Category

In [23]:
```
plt.figure(figsize=(8,5))
sns.countplot(y="category",data=df)
plt.title("Sales Count by Category")
plt.xlabel("Count")
plt.ylabel("Category")
plt.show()
```



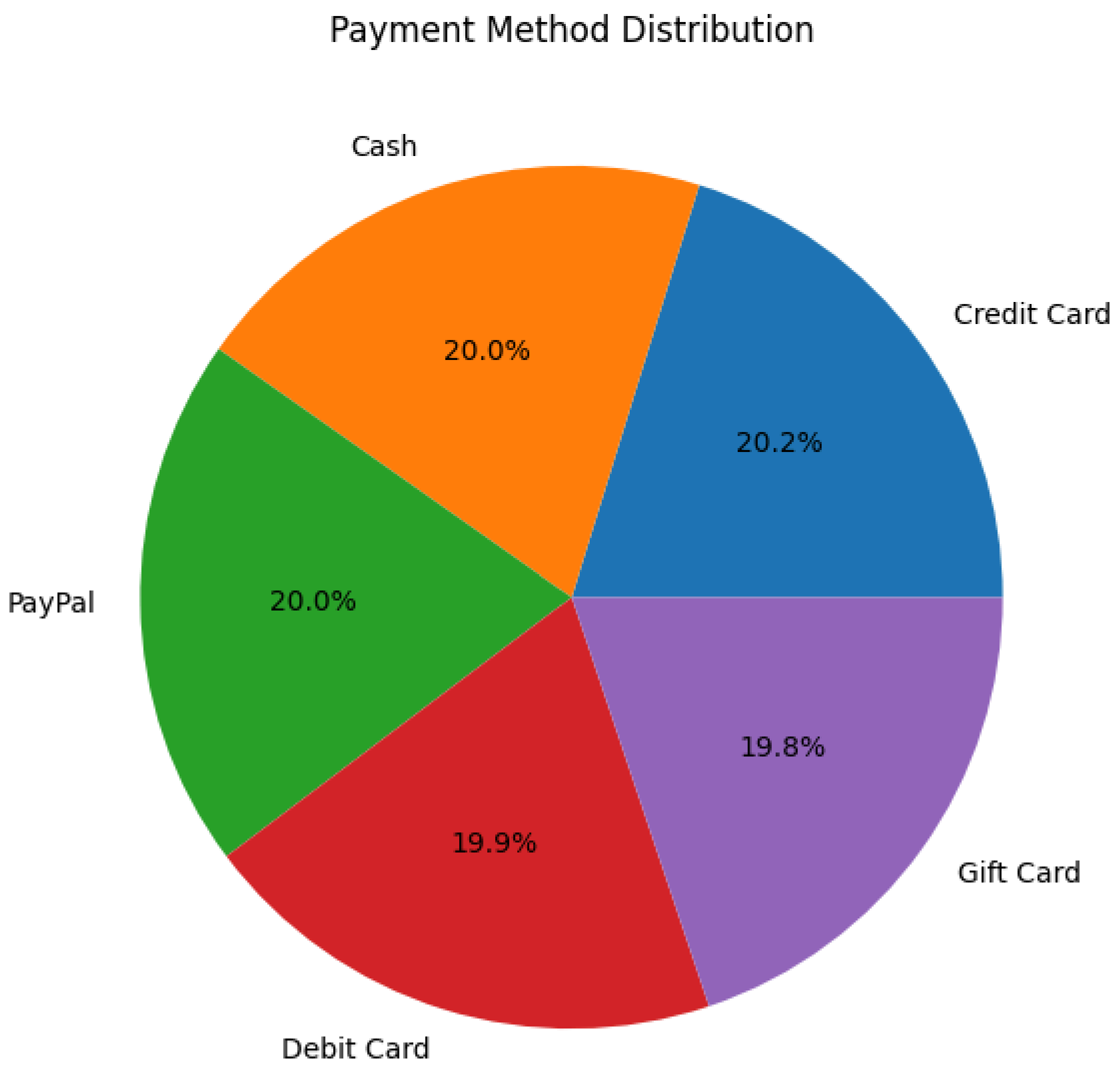
Total Sales by Region

In [33]:
```
region_sales = df.groupby('region')['sales_amount'].sum().sort_values()
plt.figure(figsize=(8,5))
plt.plot(region_sales.index,
         region_sales.values,
         marker='o')
plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.show()
```



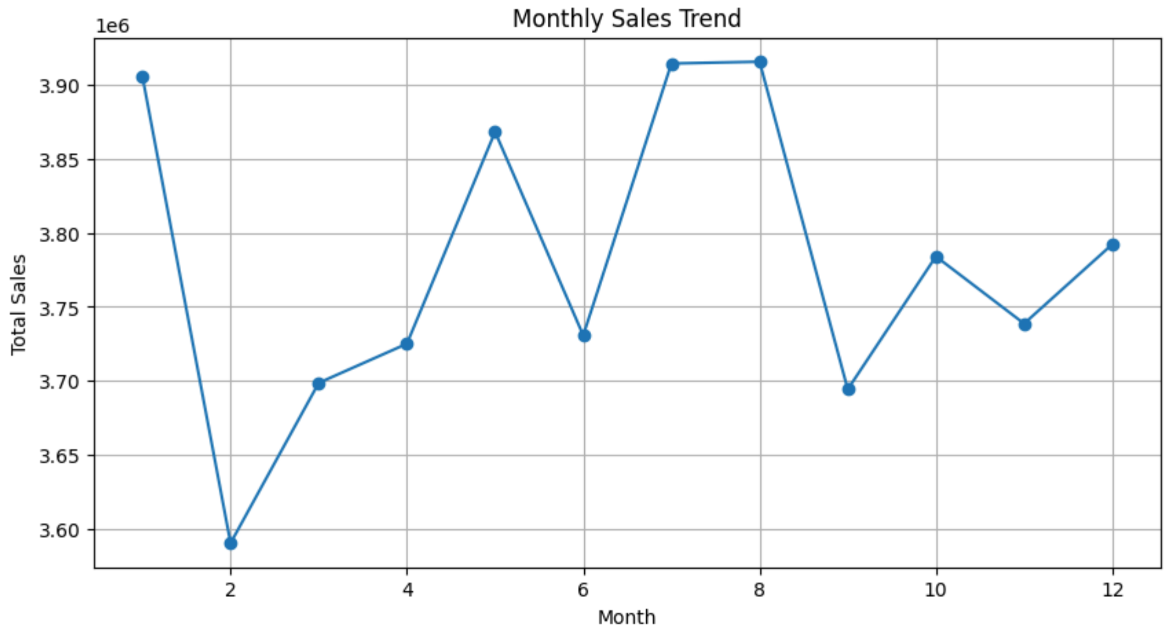
Payment Method Distribution

In [38]:
```
payment_counts=df['payment_method'].value_counts()
plt.figure(figsize=(7,7))
plt.pie(payment_counts,
        labels=payment_counts.index,
        autopct='%1.1f%%')
plt.title("Payment Method Distribution")
plt.show()
```



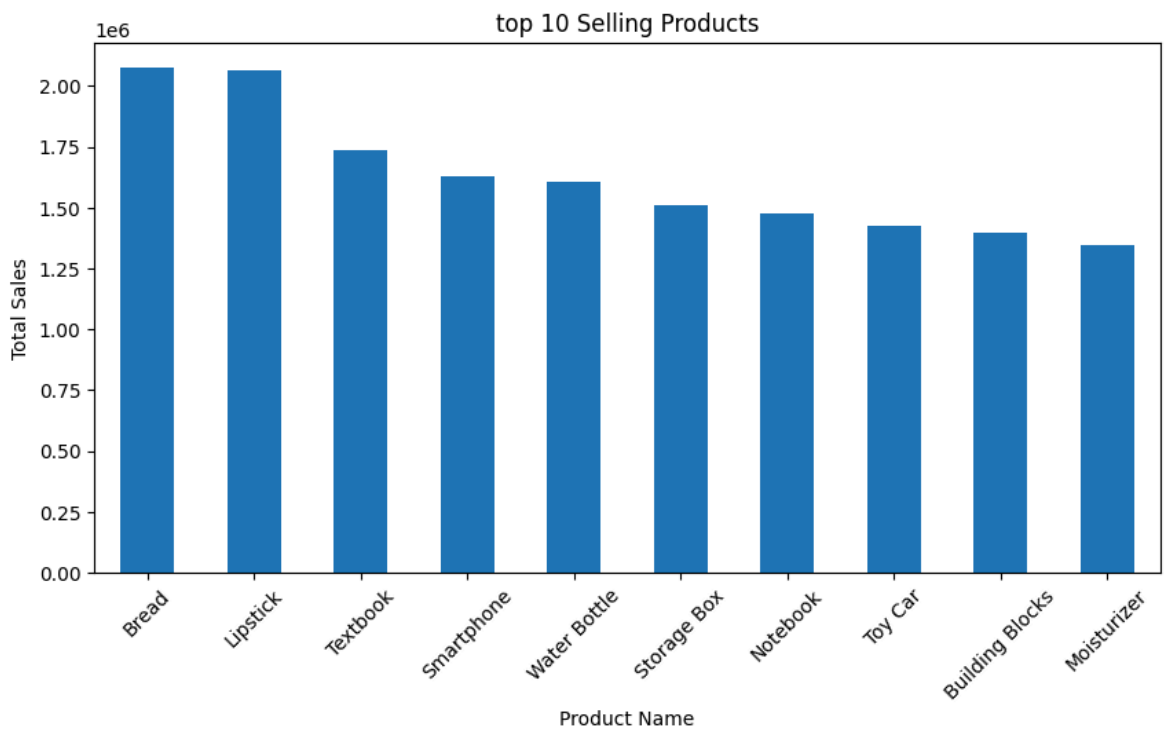
Monthly Sales Trend

In [41]:
```
df['transaction_date'] = pd.to_datetime(df['transaction_date'])
monthly_sales = df.groupby(df['transaction_date'].dt.month)['sales_amount'].sum(
plt.figure(figsize=(10,5))
plt.plot(monthly_sales.index,
         monthly_sales.values,
         marker='o')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.grid(True)
plt.show()
```



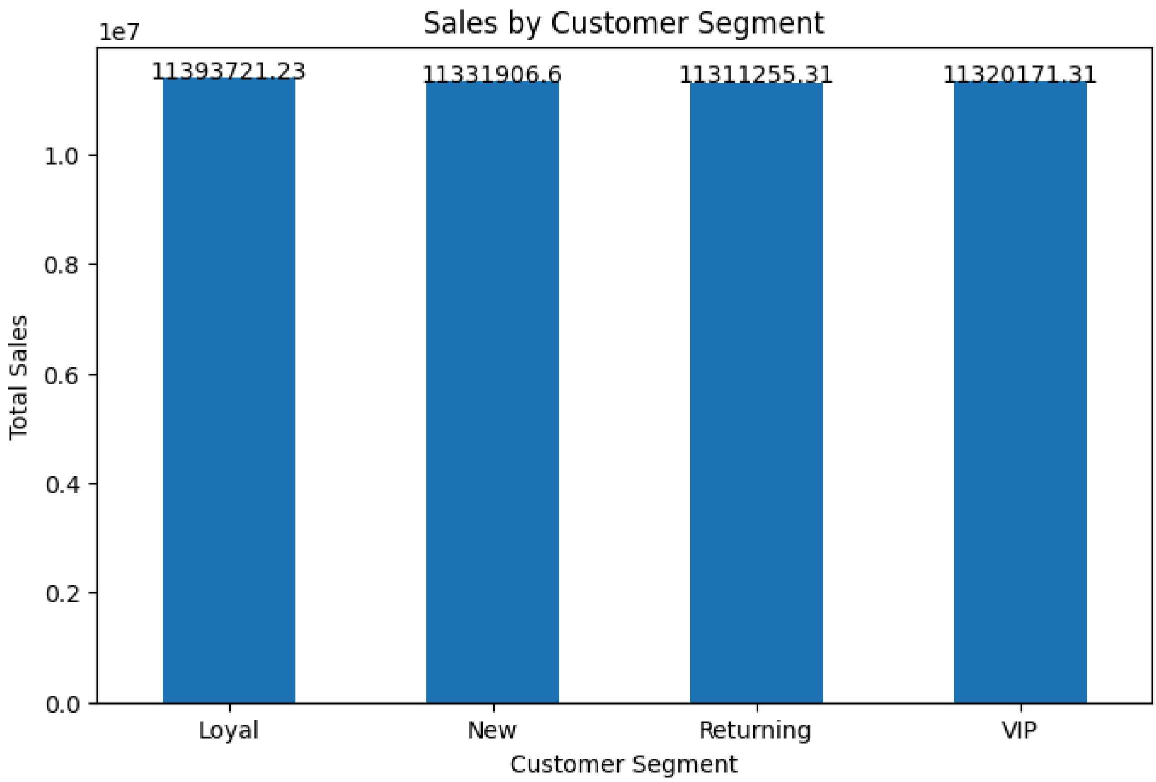
Top 10 Selling Products

In [44]:
```
top_products = df.groupby('product_name')['sales_amount'].sum().sort_values(asce
plt.figure(figsize=(10,5))
top_products.plot(kind='bar')
plt.title("top 10 Selling Products")
plt.xlabel("Product Name")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.show()
```



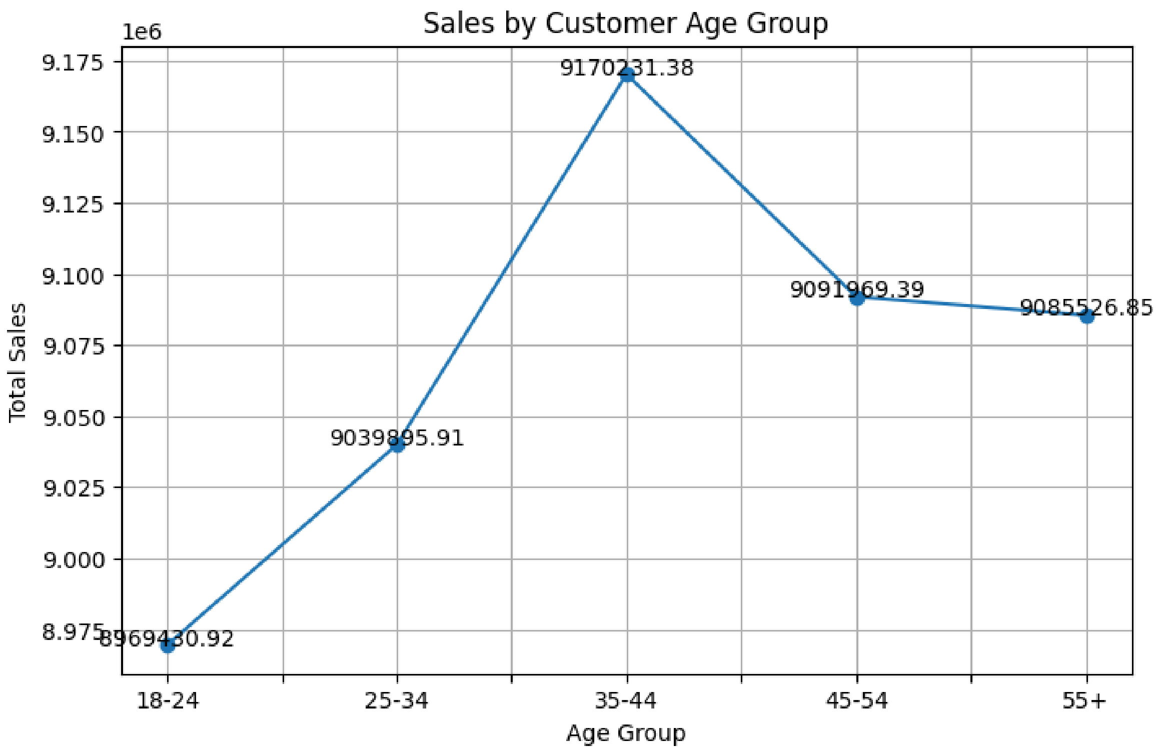
Sales by Customer Segment

In [51]:
```
segment_sales = df.groupby('customer_segment')['sales_amount'].sum()
plt.figure(figsize=(8,5))
ax=segment_sales.plot(kind='bar')
plt.title("Sales by Customer Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Total Sales")
for i , value in enumerate(segment_sales):
    plt.text(i, value, str(round(value,2)),
             ha = "center")
plt.xticks(rotation=0)
plt.show()
```



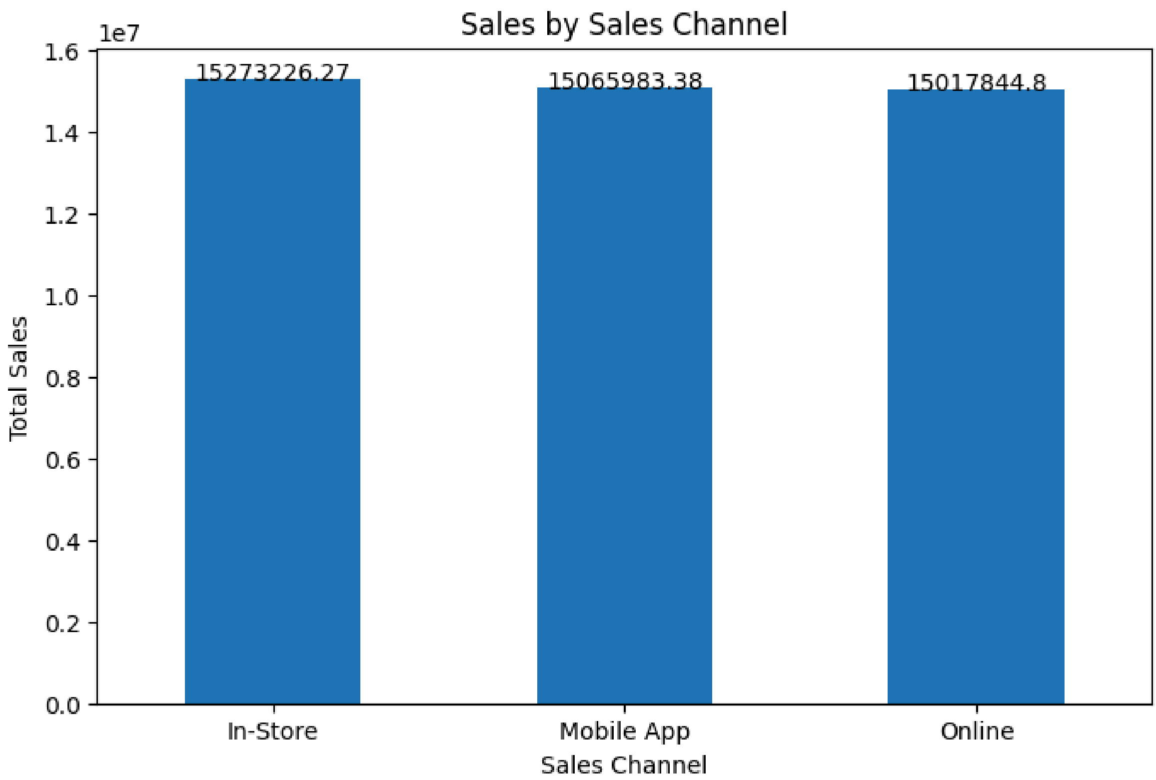
Sales by Customer Age Group

In [54]:
```
age_sales = df.groupby('customer_age_group')['sales_amount'].sum()
plt.figure(figsize=(8,5))
ax = age_sales.plot(kind='line',marker='o')
plt.title("Sales by Customer Age Group")
plt.xlabel("Age Group")
plt.ylabel("Total Sales")
plt.grid(True)
for i, value in enumerate(age_sales):
    plt.text(i, value, str(round(value,2)),
             ha = 'center')
plt.show()
```



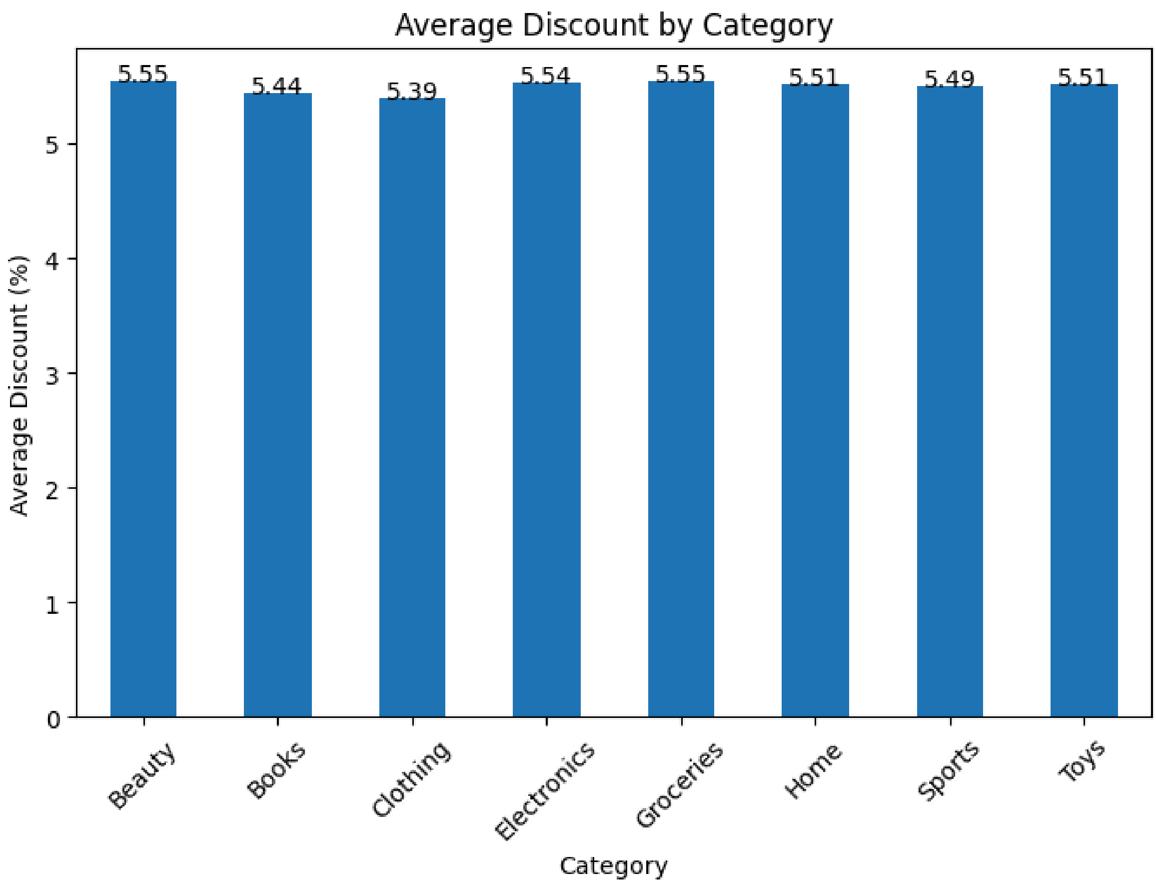
Sales Channel Analysis

In [57]:
```
channel_sales = df.groupby('sales_channel')['sales_amount'].sum()
plt.figure(figsize=(8,5))
ax = channel_sales.plot(kind='bar')
plt.title("Sales by Sales Channel")
plt.xlabel("Sales Channel")
plt.ylabel("Total Sales")
for i, value in enumerate(channel_sales):
    plt.text(i, value, str(round(value,2)),
             ha = 'center')
plt.xticks(rotation=0)
plt.show()
```



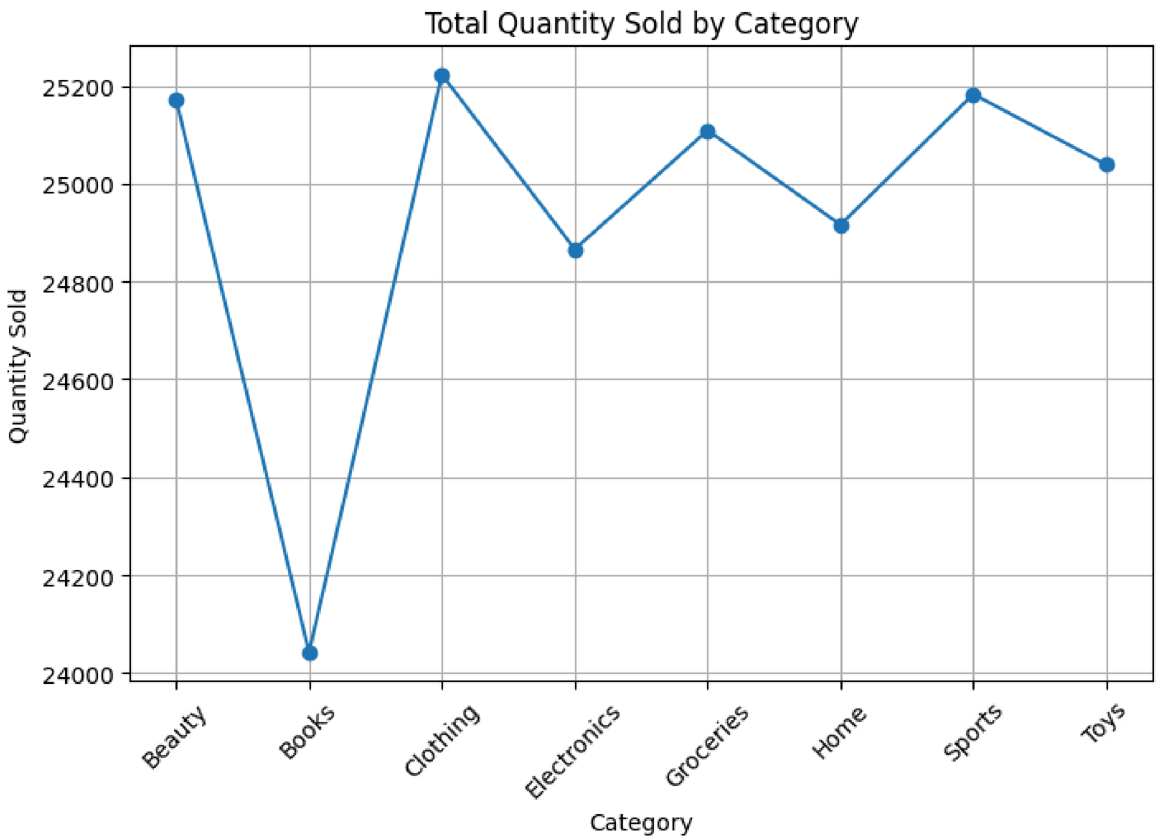
Average Discount by Category

In [61]:
```
discount_avg = df.groupby('category')['discount_pct'].mean()
plt.figure(figsize=(8,5))
ax = discount_avg.plot(kind='bar')
plt.title("Average Discount by Category")
plt.xlabel("Category")
plt.ylabel("Average Discount (%)")
for i, value in enumerate(discount_avg):
    plt.text(i, value, str(round(value,2)),
             ha='center')
plt.xticks(rotation=45)
plt.show()
```



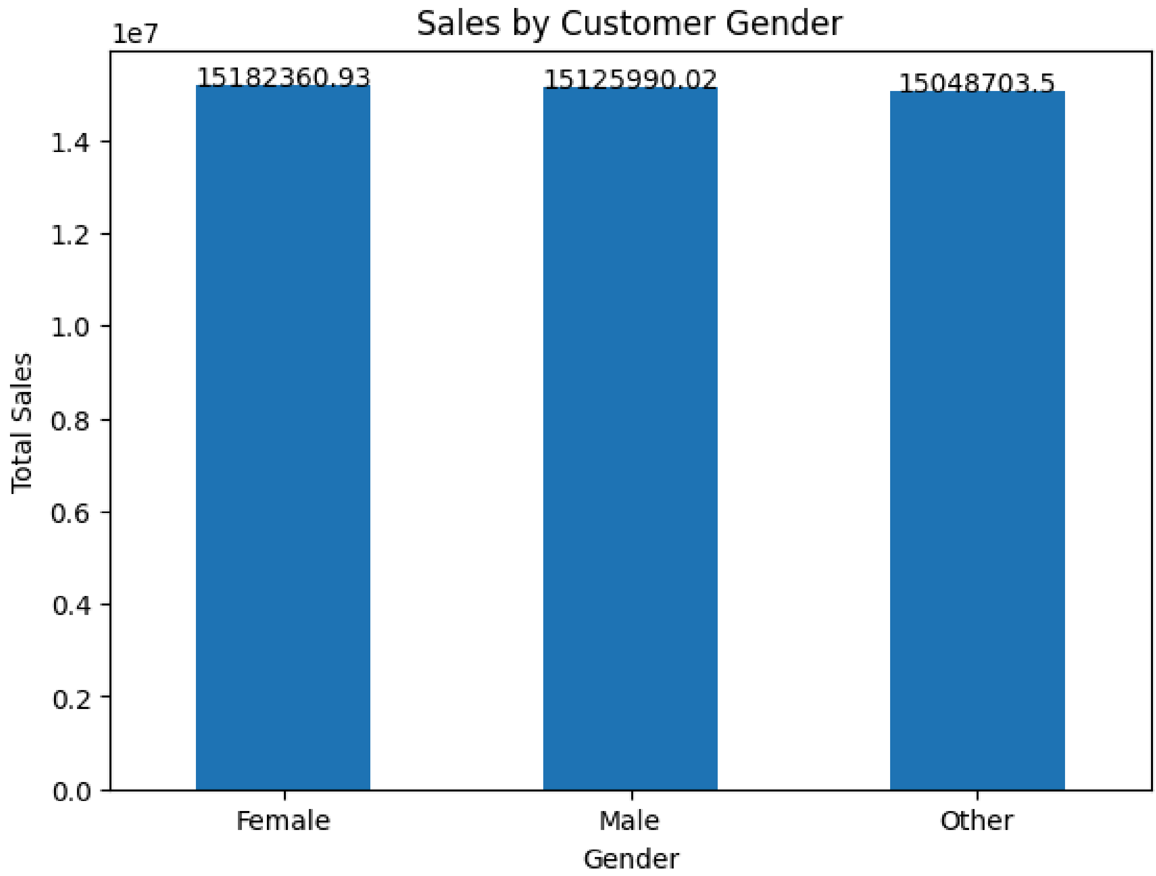
Total Quantity Sold by Category

In [64]:
```
quantity_sales = df.groupby('category')['quantity'].sum()
plt.figure(figsize=(8,5))
quantity_sales.plot(kind='line', marker='o')
plt.title("Total Quantity Sold by Category")
plt.xlabel("Category")
plt.ylabel("Quantity Sold")
plt.grid(True)
plt.xticks(rotation=45)
plt.show()
```



Sales by Customer Gender

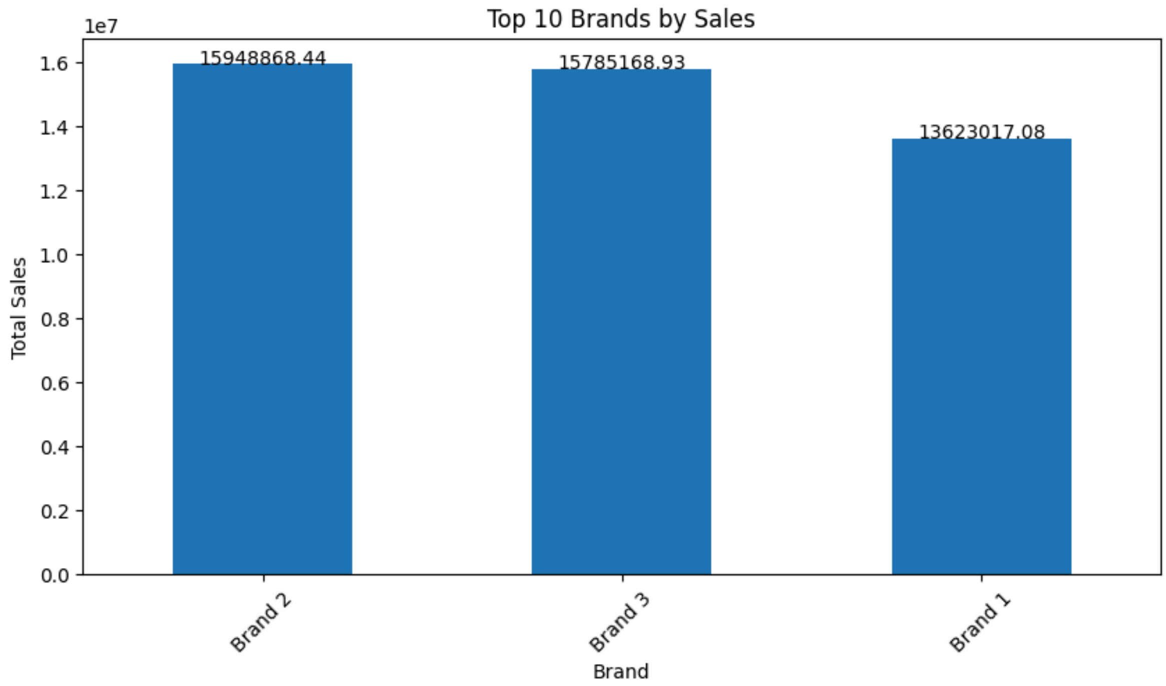
In [67]:
```
gender_sales = df.groupby('customer_gender')['sales_amount'].sum()
plt.figure(figsize=(7,5))
ax = gender_sales.plot(kind='bar')
plt.title("Sales by Customer Gender")
plt.xlabel("Gender")
plt.ylabel("Total Sales")
for i, value in enumerate(gender_sales):
    plt.text(i, value, str(round(value,2)),
             ha='center')
plt.xticks(rotation=0)
plt.show()
```



Top 10 Brands by Sales

In [70]:

```
top_brands = df.groupby('brand')['sales_amount'].sum().sort_values(ascending=Fal
plt.figure(figsize=(10,5))
ax = top_brands.plot(kind='bar')
plt.title("Top 10 Brands by Sales")
plt.xlabel("Brand")
plt.ylabel("Total Sales")
for i, value in enumerate(top_brands):
    plt.text(i, value, str(round(value,2)),
             ha='center')
plt.xticks(rotation=45)
plt.show()
```



Average Sales Amount

In [77]:
```
average_sales = np.mean(df['sales_amount'])
print("Average Sales Amount:", round(average_sales,2))
```

```
Average Sales Amount: 377.98
```

Key Insights

1. Certain product categories generated higher sales.
2. Sales performance was balanced across regions.
3. Monthly sales trends remained stable.
4. Both online and offline channels contributed well.
5. Some brands and products performed better than others.
6. Customer segments showed nearly equal contribution.
7. Discounts influenced customer purchases.
8. Different age groups actively participated in sales.

Conclusion

The Retail Sales Data Analysis project helped in understanding customer behavior, sales trends, regional performance, and product demand using Python. Different data visualization techniques were used to identify meaningful business insights. This analysis can support better business decisions and improve overall sales performance.